

TED[15] 6045

Reg. No.....

(REVISION-2015)

Signature.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE
MODEL QUESTION PAPER

RADAR AND NAVIGATION

[Time : 3 hours]

(Maximum marks: 100)

PART - A

(Maximum marks: 10)

Marks

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. What is the significance of pulse repetition frequency?
2. What is Doppler effect in radar system?
3. Define LORAN
4. List different types of satellite navigation techniques.
5. What is GNSS? (5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Briefly explain the radar performance factors.
2. Compare a pulse radar with a CW radar.
3. With the help of diagrams explain A-Scope display.
4. State the limitations of DME
5. List four methods of navigation
6. Describe space segment of GPS
7. State the basic principle of INS and DNS (5x6=30)

PART - C
(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

- III. a. With the help of a block diagram explain in detail the operation of radar (10)
- b. Briefly discuss the frequency ranges used in radar. (5)

OR

- IV. a. Derive the radar range equation. (9)
- b. Write short notes on
- (i) Signal to noise ratio (ii) Radar cross section (6)

UNIT – II

- V. With the help of block diagram explain the operation of FM CW radar. (15)
- OR

- VI. With the help of block diagram explain the operation of MTI radar. (15)

UNIT – III

- VII.a. Describe hyperbolic navigation (8)
- b. Explain the working principle of VHF phase comparison ADF (7)

OR

- VIII.a. Explain the working principle of VOR, Doppler VOR, and DME (10)
- b. Describe different types of radio range (5)

UNIT – IV

- IX. a. Explain the function of ILS (10)
- b. List the applications of GPS (5)

OR

- X. a. Describe the following:
- i. COMPASS ii. DORIS iii. GALILEO (15)